0(Please write your Enrollment Number) Enrollment No.________________

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name:<B.TECH>, Semester:<>

(December, 2024)

Subject Code: BAI 110	**Subject: PROGRAMMING WITH PHYTHON**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	**CO Mapping**
	a) Write a Python program to count the number of even and odd numbers from a series of numbers.		CO1
	b) Count number of digits in a Number. Print the digits of Number.		CO1
	c) State the difference between lists and dictionary datatypes.		CO2
	d) Write a program to check if the letter 'h' is present in the word 'technical'.		CO2
	e) Write a Python program to handle a ZeroDivisionError exception when dividing a number by zero.		CO3
	f) Write a Python program that executes an operation on a list and handles an IndexError exception if the index is out of range.		CO3
	g) Write a Pandas program to create and display a one-dimensional array-like object containing an array of data using Pandas module.		CO4
	h) Write a NumPy program to reverse an array (the first element becomes the last)		CO4
	UNIT I		**CO Mapping**
Q2	(a) With the necessary examples differentiate between while and for loops. Write a Python program that prints the Fibonacci series between 0 to 50. (b) Write a python program to find number btw 100 to 400 where each digit of a number is an even number . the number obtained should be printed in a comma-seperated sequence	(10)	CO1
Q3	Print the following pattern using a while loop. 123454321 1234321 12321 121 1	(10)	CO1
	UNIT II		**CO Mapping**
Q4	(a) Write the python program the print the given series : 22,21,23,22,24,23......... print the given series from 20 to 80Explain different file operations with examples. (b) Write a Python program to read a file line by line store it into a variable.	(10)	CO2
Q5	(a) Write a python Function to reverse a string and find the product of all the numbers in a list. (b) Write a User defined function that reads some integers b/w 1 to 20 and counts the occurrences of each. SAMPLE: Input: 2 5 2 5 8 8 10 8 2 4 4 5 2	(10)	CO2

<table>
<tr><td></td><td>Output: 2 occurs 4 times
5 occurs 3 times
8 occurs 3 times
10 occurs 1 times
4 occurs 2 times</td><td></td><td></td></tr>
<tr><td colspan="4">UNIT III</td></tr>
<tr><td></td><td></td><td></td><td>CO Mapping</td></tr>
<tr><td>Q6</td><td>(a) Explain with examples the difference between Syntax Error and Exceptions.
(b) Can multiple exceptions be handled using a single except clause? Explain with example.</td><td>(10)</td><td>CO3</td></tr>
<tr><td>Q7</td><td>(a) What are the common exceptions in Python and how to catch them?
(b) Can I use try without except in Python? Explain.</td><td>(10)</td><td>CO3</td></tr>
<tr><td colspan="4">UNIT IV</td></tr>
<tr><td></td><td></td><td></td><td>CO Mapping</td></tr>
<tr><td>Q8</td><td>Write a Pandas program to create and display a DataFrame-
<table>
<tr><th></th><th>attempts</th><th>name</th><th>qualify</th><th>score</th></tr>
<tr><td>a</td><td>1</td><td>Anastasia</td><td>yes</td><td>12.5</td></tr>
<tr><td>b</td><td>3</td><td>Dima</td><td>no</td><td>9.0</td></tr>
<tr><td>c</td><td>2</td><td>Katherine</td><td>yes</td><td>16.5</td></tr>
<tr><td>d</td><td>3</td><td>James</td><td>no</td><td>NaN</td></tr>
<tr><td>e</td><td>2</td><td>Emily</td><td>no</td><td>9.0</td></tr>
<tr><td>f</td><td>3</td><td>Michael</td><td>yes</td><td>20.0</td></tr>
<tr><td>g</td><td>1</td><td>Matthew</td><td>yes</td><td>14.5</td></tr>
<tr><td>h</td><td>1</td><td>Laura</td><td>no</td><td>NaN</td></tr>
<tr><td>i</td><td>2</td><td>Kevin</td><td>no</td><td>8.0</td></tr>
<tr><td>j</td><td>1</td><td>Jonas</td><td>yes</td><td>19.0</td></tr>
</table>
(a) Write a Pandas program to get the first 3 rows of the given DataFrame
(b) Write a Pandas program to select the rows where the number of attempts in the examination is greater than 2
(c) Write a Pandas program to count the number of rows and columns of the DataFrame.
(d) Write a Pandas program to select the rows where the score is missing, i.e. is NaN
(e) Write a Pandas program to calculate the sum of the examination attempts by the students</td><td>(10)</td><td>CO4</td></tr>
<tr><td>Q9</td><td>(a) Write a NumPy program to create a structured array from given student name, height, class and their data types.
Original array:
[(b'James', 5, 48.5) (b'Nail', 6, 52.5) (b'Paul', 5, 42.1) (b'Pit', 5, 40.11)]
(i) Sort the array on height
(ii) Sort by class, then height if class are equal
(b) Write a Python GUI program to create three single line text-box (Name, Use ID, Password) to accept a value from the user using tkinter module.</td><td>(10)</td><td>CO4</td></tr>
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